Department of Computer Centre
Rollwala Computer Centre
Gujarat University
MCA– I
Sessional – II

Subject: Mathematical Foundation **Date:19/12/2024**
Time: 1 hr. and 30 mins. **Total Marks:40**

Q.1 Answer the following. **(Any Three)** [21]

(a) Use the Gram-Schmidt process to transform the basis vectors: {(1,0,0), (3,7, -2), (0,4,1)} into an orthonormal basis.

(b) Solve the following systems of equations by the Gauss Jordan method:

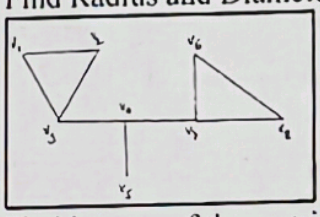
$$3x - 2y + 5z = 2$$
$$4x + y + 2z = 4$$
$$2x - y + 4z = 7$$

(c) Perform the five iterations of the bisection method to obtain a root of the equation $f(x) = x^3 - x - 1 = 0$.

(d) Perform Newton–Raphson method to find the root of $x \log_{10} x - 1.9 = 0$, correct up to three decimal places with $x_0 = 3$ and $x_1 = 4$.

Q.2 Answer the following. **(Any Three)** [12]

(a) Find Radius and Diameter of the graph G:



(b) Find inverse of the matrix

$$A = \begin{bmatrix} 4 & -2 & 3 \\ -2 & 1 & 6 \\ 1 & 2 & 2 \end{bmatrix}$$

(c) Given that $a = (\frac{4}{3}, -1, \frac{2}{3})$ and $b = (5,6,-1)$. Find the unit vector u in the direction of a. Show that b is orthogonal to a.

(d) Solve the following systems of equations by Cramer's rule.

$$3x - 2y + z = 13$$
$$-2x + y + 4z = 11$$
$$x + 4y - 5z = -31$$

Q.3 Attempt any **SEVEN** out of **EIGHT**: [7]

1. Define: K-Regular Graph
2. Define: Hand- Shaking Lemma
3. Write down the formula of False Position Method.
4. Find determinant of $A = \begin{bmatrix} 4 & 1 \\ -5 & 2 \end{bmatrix}$
5. Define: Norm of a vector
6. Define: Convex Function
7. Define: Row Echelon Form
8. Define: Trace of a Matrix

Department of Computer Science
Gujarat University
MCA theory examination
Semester I

Subject: Data Structures
Total Marks: 30 **Time: 1.5 hours**

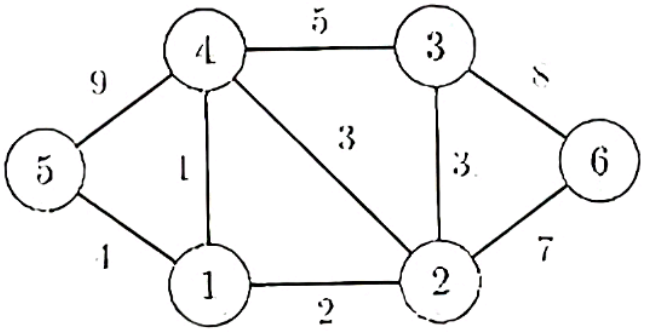
Q1. Explain the method and write algorithms for the following **(ANY TWO)**: [5]

i. Insertion before a given node in singly linked list
ii. Insertion at beginning in circular linked list
iii. Deletion of of given node in doubly linked list
iv. Deletion of last node in circular linked list

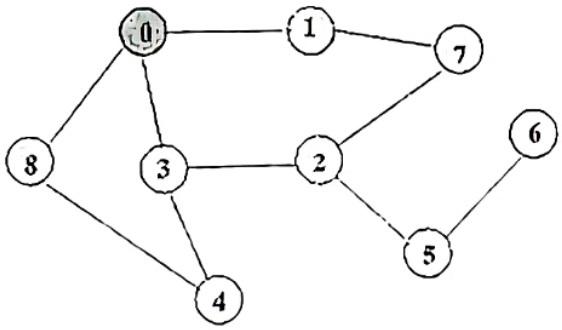
Q2. Draw expression tree for the following Prove that preorder traversal and postfix expression of expression tree is prefix and postfix equivalent of that expression [4]

i. a*(b + c) + ((d + e*f)*g)
ii. a+b/c-d*e+f^g

Q3. Find minimum spanning tree using kruskals algorithm and prims algorithm for given graph [04]



Q4. Write BFS and DFS of given graph . take 0 as start node [03]



Q5. Explain method of postfix evaluation using stack using example [5]

Q6. Write algorithm for infix to postfix conversion using stack [5]

Q7. Draw binary search tree for given data; [2]
3,7,90,12,34,56,1,2,55,100,400,2,10. Show that its inorder traversal gives ascending order

Q8. If the preorder and inorder traversal of the tree are as follows, draw the tree [2]

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow H \rightarrow I \rightarrow J \rightarrow K.$

Inorder:

Preorder:

$F \rightarrow D \rightarrow B \rightarrow A \rightarrow C \rightarrow E \rightarrow J \rightarrow G \rightarrow I \rightarrow H \rightarrow K$

DEPARTMENT OF COMPUTER SCIENCE
GUJARAT UNIVERSITY
MCA SEM I
Subject: Object Oriented Concept and Programming
Sessional Examination – II

December, 2024

Time: 60 mins
Max. Marks: 20

Answer the followings.(Any Five) **20**

1) What is inheritance? Why we need inheritance? Explain derivation used in inheritance in detail.

2) What is polymorphism? Why runtime polymorphism is important? How to achieve runtime polymorphism? Demonstrate using an example.

3) Mention the stream classes hierarchy in C++. Explain manipulators in details. How to define and use user defined manipulator?

4) Answer the followings:
   a. Differentiate ios functions and manipulators
   b. What is abstract class? How to define pure virtual function? Also Mention the use of pure virtual function.

5) It is stated that multiple inheritance should be uses very carefully. Explain the statement with justification and what to do to avoid multiple inheritance problems.

6) What is the advantage of using binary file over text file? Write syntax and example of reading and writing in binary file. Also mention how we can random access the binary file with an example.

7) What is generic programming? How template store internally? Demonstrate using an example?

*** *** ***

Department of Computer Science
Gujarat University
MCA-I
Sessional II
Subject: Problem Solving with Python
Date: 20th December, 2024

Time: 1:30 hrs

Total Marks: 20

Q1. One-mark Questions [3]

i. What are the contents of the file writing.txt created by the following snippet of code?

```
f = open('writing.txt', 'w')
f.write('one')
f.write('two')
f.write('three')
f.close()
```

a.

1	one
2	two
3	Three

b.

1	onetwothree

c.

1	one
2	
3	two
4	
5	Three

d.

1	one two three

ii. Point out the correct statement.

a. NumPy main object is the homogeneous multidimensional array
b. In Numpy, dimensions are called axes
c. Numpy array class is called ndarray
d. All of the mentioned

iii. What is the exception raised by the following snippet of code? Assume that this is a stand-alone piece of code.

```
for x in range(n):
    if x % n == 0:
        print(x)
```

a. ValueError
b. NameError
c. ZeroDivisionError
d. This program is error free.

Q2. Answer the following questions (Any Four) [12]

i. Explain the use of assert statement and user defined exception in detail.
ii. Explain the use of tell() and seek().
iii. What is inheritance?
iv. Write the syntax to read and write text file and binary file in python with example?
v. Explain array using NumPy Module?

Q3. Write a program using the below details: [5]

Write a Python class Employee with attributes like emp_id, emp_name, emp_salary, and emp_department and methods like calculate_emp_salary, emp_assign_department, and print_employee_details.

Sample Employee Data:
"ADAMS", "E7876", 50000, "ACCOUNTING"
"JONES", "E7499", 45000, "RESEARCH"
"MARTIN", "E7900", 50000, "SALES"
"SMITH", "E7698", 55000, "OPERATIONS"

i. Use 'assign_department' method to change the department of an employee.
ii. Use 'print_employee_details' method to print the details of an employee.
iii. Use 'calculate_emp_salary' method takes two arguments, salary and hours_worked, which is the number of hours worked by the employee. If the number of hours worked is more than 50, the method computes overtime and adds it to the salary. Overtime is calculated as following formula:
overtime = hours_worked – 50
Overtime amount = (overtime * (salary / 50))

Department of Computer Science
Gujarat University
MCA- I
Sessional – II

Subject: Relational Database Management Systems **Date: 16/12/2024**

Q: 1 **Define the following (Any 3)** **[06]**

a. Dirty Read
b. Functional Dependency
c. CLR
d. Shared Memory Architecture
e. WW Conflicts
f. Schedule
g. CAP in NoSQL
h. Log Tail

Q.2 **Attempt the Following (Any 3)** **[15]**

1. What is Bell La-Padula Protocol. Explain with reference to recovery
2. Discuss various transaction states with diagram
3. Explain 2PL in detail with appropriate example
4. Compare Mandatory Access Control Vs Discretionary control with appropriate examples
5. Explain Aries Recovery algorithm
6. Discuss Native Relational Algebra operators with appropriate examples
7. What is Thomas write rule. How is relevant in writing a transaction?
8. Explain Wait-Die and Wound-wait deadlock prevention technique with example

Q3 **Write Relational Algebra Queries for the following (Assume the schema)** **[04]**

a. Find names of suppliers, supplying some red part for less than 100 rupees
b. Find names of all departments that has no employees working in them
c. Find all the suppliers who supply Red and Green boat
d. Find all employees of sales department who are working for greater than 10 years.
e. Find all the suppliers who supply all parts

Q4 **Normalize the structure upto the best level possible** **[05]**

Gallery Customer History Form

Customer Name

Jackson, Elizabeth
123 – 4th Avenue
Fonthill, ON
L3J 4S4

Phone (206) 284-6783

Purchases Made

Artist	Title	Purchase Date	Sales Price
03 - Carol Channing	Laugh with Teeth	09/17/2000	7000.00
15 - Dennis Frings	South toward Emerald Sea	05/11/2000	1800.00
03 - Carol Channing	At the Movies	02/14/2002	5550.00
15 - Dennis Frings	South toward Emerald Sea	07/15/2003	2200.00

The Gill Art Gallery wishes to maintain data on their customers, artists and paintings. They may have several paintings by each artist in the gallery at one time. Paintings may be bought and sold several times. In other words, the gallery may sell a painting, then buy it back at a later date and sell it to another customer.